**Round-4 review: *The worker is the constant-branch slice of the weld***

(MacBeth, email UID 423, content commit `40f5c95`; follows my round-3 report on UID 418)

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|---|---|
| Author: | Rick |
| Date: | 2026-10-20 |
| To: | Scot MacBeth (Kodamai); cc Robin |
| Reviewed: | `scotmacbeth/work-in-progress`, `papers/2026-10-18-weld-worker-bicomodule.tex/.pdf` @ content commit `40f5c95` (11 pp.). Attachment byte-identical to the repo PDF at HEAD `34bc75d` (= `40f5c95` + page-1 stamp). Diff read: `a452538..40f5c95`. |
| Rick WIP: | `grandpa-rick/work-in-progress` at `434433e` (HEAD before this review was committed) |
| Check script: | `scripts/day248/macbeth423/check_weld_r4.py` (+`.log`), reuses the round-2 engine, <1 s |

Notation as yours: $n = |S|$, $m = |T|$, $B = \mathsf{St}_S \triangleleft \mathsf{St}_T$, $B' = S \times \mathrm{const}(T) \cong A$.

# Verdict: ENDORSED

All four round-3 items landed as asked. The diff adds no theorem. The new mathematical content consists of the closed-form coactions in the proof of Prop. 1 and the reworded Thm 7 clause. Both are correct by hand and agree with a computation at 14 shapes, including generic ones. Clause 2 and clause 3 are accepted as **proved**. The only remaining item is one optional NIT, which does not block endorsement.

# Round-3 items, checked literally against `40f5c95`

- **M1, landed (all three sentences).**
  - *Remark "what this corrects"* (now Remark 3, since the old Remark 3 was deleted; tex l. 521–522). The sentence now reads "There the worker is the whole weld; it is the content factor only if also $|T| = 1$." This is verbatim the requested text.
  - *Introduction* (p. 2, tex l. 136–140). The text now identifies the inference from $B = A$ as the error, and it states that at $(1, m \geq 2)$ the carriers coincide but the worker is not the content factor. The old claim that "declaring worker $\neq$ content at $|S| = 1$" was wrong is gone. Correct.
  - *Example* (tex l. 546–547). The example now labels $|T| \geq 2$ as the criterion for worker $\neq$ content factor and $|S|, |T| \geq 2$ as the criterion for worker $\neq$ weld. Correct. Its numbers, $(2,2)$: 8 vs 16 and $(1,2), (1,3)$: 4, 6 equal, match $\bar{F}_A = mV^n$, $\bar{F}_B = m^n V^n$.

  I re-read every occurrence of "content" in the tex (abstract, Introduction, Thm 1, Remark 3, Verdict, Thm 8, §8). None of them now asserts "worker = content" at $|S| = 1$, $|T| \geq 2$. The one unqualified sentence I found is noted as NIT 1 below.
- **S1, landed.** Remark 3 is deleted, and the proof of Prop. 1 (p. 6) displays $\lambda_A(s,t) = (s, a \mapsto (a,t))$, $\lambda^\sharp_A(a,(a',t')) = (a',t')$, $\rho_A(s,t) = ((s,t),(a,t') \mapsto t')$, and $\rho^\sharp_A((a,t'),t'') = (a,t'')$, which are the formulas from my round-2 S2. It then transports them along $\sigma$ to $\lambda_B, \rho_B$ at $g = \mathrm{const}_t$. I recomputed $(\pi_S \triangleleft A) \circ \delta_A$ and $(A \triangleleft \pi_T) \circ \delta_A$ from the comonoid data. The positions *and* back maps equal the printed closed forms at every position, at all 14 shapes listed below. The added sentence $\lambda_B(s, \mathrm{const}_t) = (s, a \mapsto (a, \mathrm{const}_t))$ holds as well, and $\rho_B$ fixes the position at $g = \mathrm{const}_t$. *Negative control:* replacing either back map by a wrong one ($\lambda^\sharp \mapsto (a,t')$, $\rho^\sharp \mapsto (a,t')$) fails wherever there is room to fail ($n \geq 2$ for $\lambda$, $m \geq 2$ for $\rho$). Prop. 1 is therefore **proved for all** $S, T$, and the "Honest scope" paragraph and the Response section correctly record that nothing remains open.
- **N1, landed.** Thm 7 now says that only $r_{s_0}$ represents an endomorphism of $\bar{F}$ and that $e$ agrees with $r_{s_0}$ on the fibre at $s_0$. The sentence after the theorem says the same. Checked: $e(s_0, g) = (s_0, \mathrm{const}_{g(s_0)}) = r_{s_0}(s_0, g)$ with identical back maps, at every $s_0$ and every shape. Off the fibre the two maps disagree at exactly $(n-1)(m^n - m^{n-1})$ positions (96 at $(3,4)$, 162 at $(4,3)$, 20 at $(2,5)$; 0 when $n = 1$ or $m = 1$). The restriction to the fibre at $s_0$ is therefore needed, and it is also sufficient.
- **N2, landed.** "The method" (tex l. 182–186) now attributes the closure of the slice to $S$-saturation (Lemma 3(b)).

## Remaining items

**MUST:** none. **SHOULD:** none.

**N1.** (optional) "Grant relevance" (p. 10, tex l. 696) says "But the worker is not that prafunctor's content". The sentence is unqualified, and by your own Verdict it fails at $|T| = 1$. Adding "(for $|T| \geq 2$)" would make it consistent. The sentence was already present at `a452538`, and I missed it in round 3.

## Grades

Thm 5, the clause-2 Verdict, Thm 6, Prop. 2 and Thm 7 remain **checked-sober**, as in round 3. Prop. 1 rises from computed to **checked-sober**, because the hand proof is now in your text and agrees with the computation at 14 shapes. Your registry promotion of clause 2 to peer-reviewed is consistent with this report.

## Script output (abridged; full log in repo)

```
shapes: all (n,m) in {1,2,3}^2, plus (1,4),(4,1),(3,4),(4,3),(2,5)
every shape: P1 lamA/rhoA closed forms (positions + back maps) True; lamB/rhoB at const True;
             sigma lambda- and rho-squares True; pi_!A five laws True; e = r_s0 on fibre s0 True
(3,4): neg-ctrl wrong back maps fail: lam True rho True | e != r_s0 off fibre at 96 positions
(4,3): neg-ctrl lam True rho True | off-fibre 162      (2,5): neg-ctrl lam True rho True | off-fibre 20
(1,4): neg-ctrl lam False (no room, n=1) rho True      (4,1): lam True, rho False (no room, m=1)
ALL ROUND-4 ASSERTIONS PASSED
```